

Exact polynomial content of an atomic singular-inner borderline de Branges–Rovnyak family

Partial result for arXiv:2405.07016

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Status: substantial partial result, likely valid. For a concrete infinite-dimensional family of extreme singular-inner symbols, the polynomials in the borderline Schur-power space are classified exactly. The source’s existential question for arbitrary extreme symbols remains open.

1. Source question

Aleman and Dahlin study generalized de Branges–Rovnyak spaces and, at the end of arXiv:2405.07016v3, ask the following question [1]. Here $\beta = m$ and $d = 1$, so the relevant kernel is

$$K_{b,m}(z, w) = \left(\frac{1 - b(z)\overline{b(w)}}{1 - z\bar{w}} \right)^m. \quad (1)$$

Furthermore, we can also consider the case when b is vector-valued. However, in this case, when $\beta = m$ Sarason’s dichotomy is only known to hold when b is a finite dimensional vector.

When $\beta > m$ all results mentioned in this remark extend to the context of several complex variables.

Question 1. If $d = 1$, $\beta = m$ and $b(z) = z$, then clearly $\mathcal{H}_\beta(b, m) = \mathbb{C}$. Does there exist $m > 1$ and b extremal in the unit ball of H^∞ such that $\mathcal{H}_\beta(b, m)$ contains all polynomials? Do they form a dense subset?

Figure 1: Question 1 on page 25 of arXiv:2405.07016v3.

In transcription: does there exist an integer $m > 1$ and an extreme point b of the unit ball of H^∞ such that $\mathcal{H}(K_{b,m})$ contains every polynomial, and, if so, are the polynomials dense? The source proves the positive conclusion for non-extreme b , but the displayed example $b(z) = z$ gives only the constants at the borderline.

2. Result and proof intuition

For $a > 0$, let

$$S_a(z) = \exp\left(-a \frac{1+z}{1-z}\right), \quad b_{N,a}(z) = z^N S_a(z), \quad N \geq 1. \quad (2)$$

The function S_a is the singular inner function associated with one atom at 1, so $b_{N,a}$ is inner and hence extreme.

Proof intuition. The model kernel for $z^N S_a$ splits into a finite polynomial block and a shifted copy of the kernel for S_a . Expanding its m -th power shows at once that all monomials through degree $m(N-1)$ occur. To show that no later monomial can be hidden in the Hilbert completion, move to the right half-plane. There S_a becomes e^{-as} , and every element of the j -th kernel power is $(s+1)^j$ times the Laplace transform of a function supported in $[0, ja]$. That Laplace transform is entire. Applied to a putative polynomial, this forces its pole at the Cayley point $s = -1$ to have bounded order, giving the same sharp degree cutoff. This settles an infinite-dimensional extreme family but does not rule out more irregular extreme non-inner symbols.

Theorem 1. For $a > 0$ and integers $N, m \geq 1$,

$$\mathcal{H}(K_{b_{N,a},m}) \cap \mathbb{C}[z] = \{p \in \mathbb{C}[z] : \deg p \leq m(N-1)\}.$$

In particular, for every $m > 1$ the infinite-dimensional space $\mathcal{H}(K_{zS_a,m})$ contains no nonconstant polynomial. In all these spaces the polynomials fail to be dense.

Proof. Write $t = z\bar{w}$ and $P_N(t) = 1 + t + \dots + t^{N-1}$. The elementary product identity for model kernels gives

$$K_{z^N S_a}(z, w) = P_N(t) + t^N K_{S_a}(z, w), \quad (3)$$

where $K_u(z, w) = (1 - u(z)\overline{u(w)})/(1 - z\bar{w})$. Consequently,

$$K_{z^N S_a}(z, w)^m = \sum_{j=0}^m \binom{m}{j} P_N(t)^{m-j} t^{Nj} K_{S_a}(z, w)^j. \quad (4)$$

The $j = 0$ term is a diagonal polynomial kernel whose coefficient is strictly positive in every degree from 0 to $m(N-1)$. Thus every polynomial of degree at most $m(N-1)$ belongs to the target space.

We next describe the other summands without an algebraic-product closure assumption. Use the Cayley coordinate

$$z = \phi(s) = \frac{s-1}{s+1}, \quad \operatorname{Re} s > 0.$$

Since $S_a(\phi(s)) = e^{-as}$, direct calculation gives

$$K_{S_a}(\phi(s), \phi(t)) = \frac{(s+1)(\bar{t}+1)}{2} \frac{1 - e^{-a(s+\bar{t})}}{s + \bar{t}}. \quad (5)$$

For $j \geq 1$,

$$\left(\frac{1 - e^{-a(s+\bar{t})}}{s + \bar{t}}\right)^j = \int_0^{ja} e^{-u(s+\bar{t})} w_j(u) du, \quad w_j = \mathbf{1}_{[0,a]}^{*j}. \quad (6)$$

The feature-map construction of the RKHS in (6) shows that every $f \in \mathcal{H}(K_{S_a}^j)$ has the form

$$f(\phi(s)) = (s+1)^j F(s), \quad (7)$$

up to a fixed nonzero scalar, where

$$F(s) = \int_0^{ja} h(u) e^{-su} w_j(u) du \quad \text{for some } h \in L^2(w_j(u) du).$$

In particular F is entire: compact support and Cauchy–Schwarz give locally uniform convergence of the integral on all of $\mathbb{C}$. This feature-space description covers the completed RKHS, not only finite products.

Expand P_N^{m-j} in (4). The standard finite-sum rule for RKHSs [2] implies that any polynomial $p \in \mathcal{H}(K_{b_{N,a},m})$ admits a decomposition

$$p = p_0 + \sum_{j=1}^m \sum_{\ell=0}^{(m-j)(N-1)} z^{Nj+\ell} f_{j,\ell}, \quad (8)$$

where $\deg p_0 \leq m(N-1)$ and $f_{j,\ell} \in \mathcal{H}(K_{S_a}^j)$. Put $q = p - p_0$. Every term on the right is divisible by z^N , so $q = z^N h$ for a polynomial h . Substituting (7) into (8) and dividing by z^N yields

$$h(\phi(s)) = \sum_{j,\ell} (s-1)^{r_{j,\ell}} (s+1)^{j-r_{j,\ell}} F_{j,\ell}(s), \quad r_{j,\ell} = N(j-1) + \ell. \quad (9)$$

Set $R = m(N-1) - N$. For every index in (9),

$$r_{j,\ell} - j = (N-1)j - N + \ell \leq m(N-1) - N = R.$$

It follows that $(s+1)^R h(\phi(s))$ extends to an entire function. If the polynomial h is nonzero of degree d , then $h((s-1)/(s+1))$ has a pole of exact order d at $s = -1$. Hence $R < 0$ forces $h = 0$, whereas $R \geq 0$ forces $d \leq R$. In either case,

$$\deg q = N + \deg h \leq N + R = m(N-1)$$

whenever $q \neq 0$. The same degree bound already holds for p_0 , proving the claimed equality.

Finally, S_a is not a finite Blaschke product, so its model space is infinite-dimensional; (3) and (4) then make $\mathcal{H}(K_{b_{N,a},m})$ infinite-dimensional. Its polynomial subspace is finite dimensional by the proved classification and therefore cannot be dense. $\square$

3. Verification, scope, and novelty

The proof was audited at three points: the positive-kernel expansion (4), the compactly supported feature representation (6), and the maximal exponent $r_{j,\ell} - j = R$. The last identity was also checked mechanically over a grid of integer parameters by `code/exponent_audit.py`. Two separate finite-section programs tested an infinite Blaschke candidate and an explicit non-inner extreme outer candidate; both showed rapidly growing lower bounds for missing monomial norms. Those numerical diagnostics are included in `code/`, but they are not used in Theorem 1.

This is a partial result only. It gives exact negative answers for a natural atomic singular-inner family, including infinite-dimensional examples, but does not decide whether some other extreme b makes all polynomials belong. The strongest attempted extension to every inner symbol produces an antiunitary conjugation on finite product sums, but identifying its extension with the formal boundary conjugate is blocked by the lack of boundary control from the ambient weighted Bergman norm.

The run indexes contained no duplicate. Bounded searches through 13 August 2026 covered the exact question text, the paper title and authors, current arXiv versions, the 2025 JFA publication, the first author’s 2026 thesis, and the higher-order de Branges–Rovnyak literature. No resolution of Question 1 or this atomic singular-inner classification was found. Novelty outside the run is not asserted pending expert review.

Human-review recommendation. Check the RKHS finite-sum decomposition in (8), the feature-space surjectivity used in (7), and the pole-order argument when $R = -1$. Preserve the partial scope: the result supplies a sharp negative family, not a solution of the existential question.

References

- [1] A. Aleman and F. Dahlin, *Generalized de Branges–Rovnyak spaces*, J. Funct. Anal. 288 (2025), no. 11, 110860; arXiv:2405.07016.
- [2] N. Aronszajn, *Theory of reproducing kernels*, Trans. Amer. Math. Soc. 68 (1950), 337–404.